Q1 **Union of Sets** 5 marks

Write a Python program to find the union of two sets.

Input:

A={1,2,3}

B={3,4,5}

ouput:

{1,2,3,4,5}

Your Code

```
A = {1, 2, 3}
B = {3, 4, 5}

print(A | B)
```

Input

stdin input

Output

```
{1, 2, 3, 4, 5}
```

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192572068RD

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Q2 **Intersection**

Write a Python program to find the intersection of two sets.

Input:

A={1,2,3}

B={2,3,4}

ouput:

{2,3}

Your Code

```
A = {1, 2, 3}
B = {2, 3, 4}
print(A.intersection(B))
```

Input

stdin input

Output

```
{2, 3}
```

Q3 **Difference**

5 marks

Write a Python program to find the difference of two sets.

Input:

A={1,2,3}

B={2,3,4}

ouput:

{1}

Your Code

```python
A = {1, 2, 3}
B = {2, 3, 4}

print(A - B)
```

Input

stdin input

Output

```
{1}
```

Visible Test Cases

Test Case 1

5.00 / 5 marks

Q4 **Symmetric Difference**

Write a Python program to find the symmetric difference of two sets.

Input:

A={1,2,3}

B={2,3,4}

ouput:

{1,4}

Your Code

```
A = {1, 2, 3}
B = {2, 3, 4}

print(A ^ B)
```

Input

stdin input

Output

```
{1, 4}
```

Q5 Subset

5 marks

Write a Python program to check whether one set is a subset of another.

Input:

A={2,3}

B={1,2,3,4}

ouput:

Subset

Your Code

```
A = {2, 3}
B = {1, 2, 3, 4}

if A.issubset(B):
    print("Subset")
else:
    print("Not a Subset")
```

Input

stdin input

Output

```
Subset
```

Visible Test Cases

Q6 **Superset**

5

Write a Python program to check whether one set is a superset of another.

Input:

A={1,2,3,4}

B={2,3}

ouput:

Superset

Your Code

```
A = {1, 2, 3, 4}
B = {2, 3}

if A.issuperset(B):
    print("Superset")
else:
    print("Not a Superset")
```

Input

stdin input

Output

```
Superset
```

Visible Test Cases

Test Case 1

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192572068RD

Q7 Disjoint Sets

5 marks

Write a Python program to check whether two sets are disjoint.

Input:

A={1,2}

B={3,4}

ouput:

Disjoint

Your Code

```python
A = {1, 2}
B = {3, 4}

if A.isdisjoint(B):
    print("Disjoint")
else:
    print("Not Disjoint")
```

Input

stdin input

Output

```
Disjoint
```

Visible Test Cases

Test Case 1 ✔

5.00 / 5 marks

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Q8 Remove Duplicates

Write a Python program to remove duplicate elements from a list using a set.

Input:

List: 1 2 2 3 3 4

ouput:

{1,2,3,4}

Your Code

```
lst = [1, 2, 2, 3, 3, 4]

result = set(lst)

print(result)
```

Input

stdin input

Output

```
{1, 2, 3, 4}
```

Visible Test Cases

Test Case 1

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Back

Q9 Common Elements

5 marks

Write a Python program to find the common elements in three sets.

Input:

A={1,2,3}

B={2,3,4}

C={2,5}

ouput:

{2}

Your Code

```
A = {1, 2, 3}
B = {2, 3, 4}
C = {2, 5}

print(A & B & C)
```

Input

stdin input

Output

```
{2}
```

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Q10 **Missing Numbers**

5 marks

Write a Python program to find the missing numbers from 1 to 10 using sets.

Input:

Given={1,2,4,5,7,10}

ouput:

{3,6,8,9}

Your Code

```
given = {1, 2, 4, 5, 7, 10}

all_numbers = set(range(1, 11))

print(all_numbers - given)
```

Input

stdin input

Output

```
{8, 9, 3, 6}
```